

A Fully-distributed Advanced Traffic Management System based on Opportunistic Content Sharing

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Abstract—Urban mobility has become one of the most challenging issue in urban centers. As a consequence, traffic congestion has become a daily problem. Several Advanced Traffic Management Systems (ATMS) have been proposed to improve overall traffic efficiency. However, these systems inefficiently exchange traffic information, which can lead to network overload. In order to overcome the mobility problem and improve the efficiency in dealing with vehicle traffic, this paper introduces a fully-distributed advanced traffic management system based on opportunistic content sharing, named PANDORA. Simulation results indicate that PANDORA outperforms the assessed solutions in various scenarios, considering different key requirements of ATMS.

I. INTRODUCTION

Traffic congestion has become a daily concern in large cities around the world. It results from the explosive population growth and the increase in the number of automotive vehicles, surpassing the transportation infrastructure capacity [1], [2]. In this way, beyond traffic congestion, other concerns arise including worsening quality of life, negative effects in regional economic and unfriendly environments [1].

In order to minimize traffic congestion and reduce its negative impact, ATMS have been proposed, improving the efficiency of available transportation infrastructure, and integrating sensing, communication, and data-processing technologies [2]. In other words, ATMS exploit traffic-related data provided by vehicles, traffic lights, in-road and roadside sensors to identify or forecast traffic hazards.

The basis for the deployment of an efficient ATMS are Vehicular Ad hoc Networks (VANETs), which are mobile ad hoc networks composed of vehicles with embedded sensors, processing units, and wireless communication interfaces [1]. Thanks to these features, vehicles can communicate among themselves to create an ad hoc network, or they can communicate with roadside infrastructure (RSU) to increase the network capacity, providing better management as well as Internet access to the vehicles. In summary, VANETs enable information exchange among vehicles and infrastructure, and also provides a sensing and actuation platforms for ATMS [2].

Several ATMS have been proposed to deal with mobility needs and to improve the efficiency in dealing with traffic [3], [4], [5], [6], [7], [8]. On one hand, some solutions [5], [6], [3], [7] rely on centralized approach to detect congested roads as well as to suggest alternative routes. However, these solutions are not scalable and they can introduce overhead especially in scenarios with high density of vehicles such as those spanning rush hours. On the other hand, some solutions [4], [9] employ a distributed approach, but they do not have a precise knowledge about the traffic condition, which incurs in misleading route guidance.

Due to the shortcomings of current centralized approaches such as lack of scalability, distributed approach becomes quite appealing for the deployment of efficient traffic management systems. However, the key challenge in these systems is how to provide precise knowledge about the traffic condition to all

vehicles since straightforward solutions can potentially overload the network.

In order to overcome such problem, we introduce PANDORA (Preventing traffic congestion through a fully-Distributed Rerouting Algorithm), a fully-distributed ATMS designed to improve traffic efficiency, which does not overload the network. PANDORA prioritizes the provisioning of knowledge about traffic condition in critical areas (areas with frequent traffic hazards) instead of providing overall knowledge about traffic conditions. Furthermore, PANDORA employs a *Floating Content* approach, which acts as location-based service for disseminating geo-localized information instead of spreading information over the whole network [10], [11]. In general, a floating content is attached to a geographical area and it is kept available to vehicles within that area as well as to vehicles that will arrive in the area, by using opportunistic content sharing. In this way, upon approaching critical area or within it, vehicles will receive a precise information about the traffic condition, allowing them to detect congested roads and to compute alternative routes.

We present an extensive set of experiments that shows the need for novel advanced traffic management systems to improve traffic efficiency, which clearly indicates the superior performance of PANDORA when compared to existing proposals.

This paper is organized as follows. Section II provides an overview of the existing ATMS and presents related work. Section III describe PANDORA. Performance evaluation of PANDORA is presented in Section IV. Section V draws conclusions and indicates future work.

II. RELATED WORK

Some ATMS use a centralized approach to provide better traffic management [3], [5], [6], [12], [7]. These ATMS rely on detecting and suggesting alternative routes for vehicles based on information periodically supplied by them. Differently, other ATMSs use a cooperative approach [9], [4] which neither rely on any infrastructure nor detect congested roads as well as compute alternative routes. However, centralized ATMS can overload the network when the density of vehicles is high [3], [5], [6], [8], [12], [7] in case they do not employ any mechanism to reduce the number of transmissions. Distributed ATMS can incur in the same problem due to inefficient information exchange between vehicles. Additionally, they do not have a precise knowledge about the traffic condition [9], [4].

Meneguette et al. [4] introduced an ATMS named UCONDES, which uses an Artificial Neural Network to detect and classify congestion levels based on the average speed and the density of vehicles on them. These information is obtained via beacons periodically sent by all vehicles. After classifying the target road, the information is sent to other vehicles and they can decide to keep their current route or compute an alternative one. However, the mechanism to alert the vehicles about the congested roads can overload the network. As a consequence,

several vehicles in the same road can produce the same road classification and disseminate redundant messages.

Brennand et al. [5] proposed a centralized ATMS to optimize the flow of vehicles. It is spread over a set of several RSUs. Each RSU gathers data from vehicles within its coverage and builds knowledge about the traffic condition within its coverage area. Furthermore, every RSU periodically re-routes all vehicles within its coverage area using a k Shortest Paths algorithm. However, different than UCONDES, this approach has better knowledge about the traffic condition, and can compute better alternative routes, but it can introduce high overhead in dense networks, since it is centralized.

In previous work [6], we introduce SCORPION, a centralized ATMS that classifies the traffic condition based on the K-NN algorithm. Upon detecting congested roads, SCORPION employs a cooperative re-routing algorithm, which reroutes only the vehicles that will pass through a congested road. This cooperative approach balances the traffic over all roads and reduces the overall time spent in congestion. However, as the work in [5], [3], it can produce high loads in dense networks.

The above mentioned proposals have some limitations such as overloading the network [3], [5], [6], [8], [7], absence of precise knowledge about traffic condition to provide a better route guidance [9], [4]. In this way, to overcome such limitations, we introduce a fully-distributed ATMS based on opportunistic content sharing, which neither overloads the network nor introduces high overhead. In addition, it provides precise knowledge about the traffic condition in critical areas.

III. PROPOSED SOLUTION

PANDORA is a floating content based ATMS to improve overall traffic efficiency. It employs an opportunistic content sharing to provide and store information about traffic condition within critical areas. It does not use any infrastructure, consequently enable vehicles to detect congested road and improve their routes.

Definition: Consider a VANET environment represented by a simple directed and weighted graph $G = (V(G), E(G))$, where the set $V = \{v_1, v_2, \dots, v_i\}$ corresponds to the set of intersections (vertices), the set $E = \{e_{01}, e_{12}, \dots, e_{ij}\}$ corresponds to the set of roads (edges) which connects the intersections $E(G) \subseteq V(G) \times V(G)$, where the edge e_{ij} is defined by a pair of subsequent vertices $(v_i, v_j) \in V(G)$. Moreover, $W = \{w_{01}, w_{12}, \dots, w_{ij}\}$ is a set of weights representing the traffic condition, in which $w : E \rightarrow \mathbb{R}_+^*$. Let $N = \{n_1, n_2, \dots, n_i\}$ be a set of vehicles (nodes) and $R = \langle e_{ij}, \dots, e_{mn} \rangle \mid R \subseteq E(G)$ be a route $\forall n \in N$.

A key concern in providing a fully-distributed ATMS is how to build and deliver to all vehicles precise knowledge about the traffic condition of the entire scenario producing low overhead and without overloading the network. The information of each vehicle needs to be spread to each other, and if it is not done in an efficient way, several transmissions in a short period of time can overload the network, introducing undesired overhead [1]. In this way, vehicles will not have precise knowledge about the traffic condition, incapacitating them to detect congested roads. To deal with this concern, PANDORA builds and delivers knowledge about traffic condition just for critical areas i.e., areas which have known locations and known characteristics such as traffic hazards, incidents, traffic congestion and so on. Moreover, it uses an opportunistic content sharing to provide knowledge as a floating content within the critical area. Consequently, whenever a vehicle arrives at each critical area, it

will receive the floating content, enabling it to detect congested roads, and it will check whether its current route will pass through a congested road. In this case, making necessary the discovery of a better route.

Critical areas have location (x_c, y_c) and size C , which define its center (critical location) and the area potentially affected by the events within it. For each floating content PANDORA attributes an *Anchor Zones* (AZ) defined by a radius $r_a \in \mathbb{R}_+^* \mid r_a \geq \frac{C}{2}$ which is based on the critical location (x_c, y_c) . Furthermore, each AZ is responsible to hold the floating content within its area making it available not only to all vehicles within it, but also to the vehicles that will arrive. They may have different sizes and can be set up for different purposes [11]. Moreover, they can know locations i.e., they may take place as soon as a traffic hazard or incident occurs.

Figure 1 describes the AZ for two floating content. In particular, AZs (Az_1 and Az_2) have different sizes and their respective floating content stored. Vehicles n_1 and n_2 have the floating content provided by Az_1 and vehicles n_4 and n_5 have the floating content provided by Az_2 . Therefore, if a vehicle is in an intersection of two or more AZ, it potentially can have the floating content provided from more than one AZ, which is the case of vehicles n_3 and n_6 that have the floating content provided by Az_1 and Az_2 .

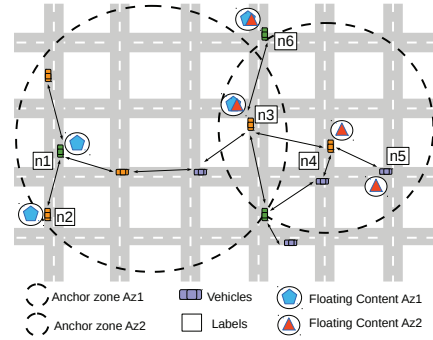


Fig. 1. Floating content and their respective Anchor zones.

Despite the critical areas known by PANDORA, it does not have any knowledge about the traffic condition within them. In this way, it employs an efficient mechanism to build precise knowledge about the traffic condition within these areas, which is further available as a floating content ensuring an opportunistic content sharing and its survivability. Thereafter, it also provides a mechanism to detect congested roads to every vehicle, enabling them to have a better route guidance. Subsection III-A describes how these floating content are built, Subsection III-B describes how PANDORA ensures the availability and survivability of each floating content, and Subsection III-C describes how vehicles detect congested roads and compute alternative routes to avoid congested roads.

A. Floating Content Creation

Building a precise knowledge about the traffic condition relies on periodic information shared by all vehicles through beacon messages, which contains their current position $e_{ij} \in R$, average speed s_m , direction and an identifier to inform if it updated floating content is stored.

Given a critical location (x_c, y_c) and its area C , PANDORA defines a set of *source vehicles* $N' \subseteq N$, which are within

a critical area and which are responsible for building the knowledge about the traffic condition in that area. However, there are some concerns about (i) how to ensure the whole coverage of the critical area, once the communication range of the vehicles is pretty smaller than the critical area, and (ii) how to build this knowledge without overloading the network.

In this way, PANDORA segments the whole coverage of the critical area into k smaller areas called *subareas*, to first build knowledge about the traffic condition within each subarea, then spread the knowledge of each subarea to each other, enabling an aggregation of the knowledge of all k -subareas to build the floating content. The number of k -subareas is based on [9].

Before building knowledge of each subarea, first each source vehicle $n_i \in N'$ builds local knowledge about the traffic condition by aggregating beacons received from its neighbors. This local knowledge is represented by the set $E^{n_i} = \{e_{ij}, \dots, e_{mn}\} \mid E^{n_i} \subset E(G)$, which is composed by the roads within its coverage and the respective average speed. Upon creating its local knowledge E^{n_i} , each source vehicle shares it with its neighbors in the same subarea in order to create knowledge about the traffic condition of the whole subarea. Therefore, the aggregation function is given by

$$S_i = \frac{1}{n} \sum_{n_i \in N'} e_{ij}, \forall e_{ij} \in E^{n_i} \quad (1)$$

where, S_i with $i \in \{1, 2, \dots, k\}$ represents the knowledge about the traffic condition of each subarea composed by the aggregation of the knowledge E^{n_i} received by all source vehicles within it. In addition, based on the average speed of each road, each vehicle sets up the road weight using

$$w_{ij} = 1 - \max(e_{ij})^{-1} \times e_{ij} \quad \forall e_{ij} \in E^{n_i} \quad (2)$$

where $\max(e_{ij})$ and e_{ij} denote the max speed, and the average speed of each road.

Each source vehicle in the same subarea builds an edge-induced subgraph $G[S_i] \subset G$ with the aggregated knowledge S_i , where the set $E(G[S_i]) \subset E(G)$ represents the roads within each subarea, and the set $V(G[S_i]) \subset V(G)$ represents the set of vertices that connects each road $e_{ij} \in E(G[S_i])$. In this way, to build the floating content with the knowledge about the traffic condition of the whole critical area, each edge-induced subgraph $G[S_i]$ needs to be aggregated into a common one. Therefore, each subgraph of each subarea has to be forward to each other. PANDORA employs a broadcast-oriented inter-subarea dissemination, so that it does not overload the network within the critical area, such decision is based on [1] which employs an efficient broadcast suppression mechanism and reduces the number of packet collisions cancelling redundant transmissions, avoiding the synchronization introduced by the IEEE 802.11p standard.

Thanks to the inter-subarea dissemination jointly with segmentation of the critical area into smaller subareas, PANDORA maximizes the coverage reducing the number of transmissions and ensuring at least a single vehicle within each subarea. As each subarea is distinct, transmission in one subarea does not cancel the transmission in another one. However, if such segmentation were not employed, this approach would create gaps in the overall knowledge, since the transmission of a vehicle would potentially cancel the transmission of many others sharing the same region [1]. Information about those vehicles can be pretty important to provide precise knowledge about the traffic condition. Furthermore, such inter-subarea dissemination process does not overload the network [1].

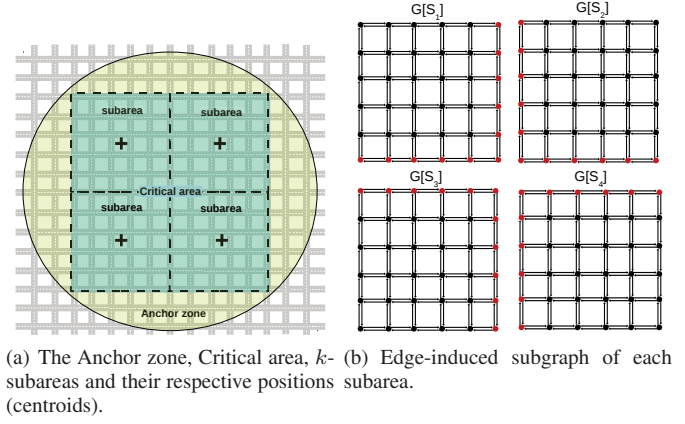


Fig. 2. Critical areas, subareas, Anchor zones and edge-induced subgraphs.

After receiving every subgraph of each subarea, PANDORA creates the floating content which contains a subgraph $G' \subseteq G$ given by the Equation below which represents the whole critical area, an anchor radius r_a to determine its geo-location, a timestamp to inform when the floating content was created and an identifier to identify which critical area such floating content regards to.

$$G' = \bigcup_{i=0}^k G[S_i] \quad (3)$$

For the sake of clarity, Figure 2 describes each subarea of a critical area, with the respective subgraph created by PANDORA. In particular, Figure 2(a) shows the segmentation employed by PANDORA showing the Anchor zone, critical area, k -subareas and their respective positions i.e. centroids calculated based on [9]. In addition, Figure 2(b) shows each edge-induced subgraph $G[S_i]$ created in each subarea. Moreover, the red vertices represents the common vertices of two subgraphs i.e. the vertices that are used to join all subgraphs into a single one to represent the whole critical area. It is important, to stress that in the join process, multiple edges are removed to result in a simple graph as well.

B. Opportunistic Content Sharing

The Opportunistic Content Sharing is essential to ensure the floating content survivability on its AZ. It holds each floating content within its AZ, making it available for vehicles in the critical area and vehicles that will arrive in this area. In addition to the AZ, PANDORA employs another zone called *Forwarding Zone* (FZ), which has radius $r_f \in R_+^* \mid r_f > r_a$ and which is responsible to send back the floating content stored in vehicles leaving the AZ. In other words, vehicles which have moved out from the AZ but still store the floating content needed to seeks actively for vehicles that are approaching within the AZ to forward the floating content.

To avoid unnecessary transmissions, when a vehicle that stores a floating content leaves its FZ, it discards the floating content to avoid forwarding it to vehicles that are far away, since they do not have interest on it at this moment, and the traffic condition of such critical area in the floating content has possibly been changed when they arrive in it. In addition, as soon as they approach critical areas, they will receive the floating content that regards to these critical areas with an updated information about traffic condition.

The forwarding process ensure the survivability of the floating content in its AZ and it is based on the interest that a vehicle has on such floating content. In summary, the interest depends on whether it will pass through a critical area, and if they store some floating content that are not up to date. This information is obtained through beacon messages periodically shared by all vehicles. Therefore, when a vehicle storing an up to date floating content receives a beacon from another vehicle, and they are within the AZ or FZ, it will forward the floating content based on the probability p

$$p = \begin{cases} 1 & \text{if } \exists e_{ij} \in R \mid e_{ij} \in E(G') \\ \theta(R) & \text{Otherwise} \end{cases}$$

where, $\theta(R) \in [0, 1]$ is some decreasing function that gives the forwarding probability when vehicles will not pass through the AZ.

C. Congestion detection and route guidance

The floating content available enables vehicles that have such content to detect congested roads and improve their route guidance, computing alternative routes to avoid congested road. Each vehicle can detect congested roads based on their weight w_{ij} , $\forall e_{ij} \in E(G')$, and it can classify the traffic condition in those roads. The classification is based on the Level-Of-Service (LOS) of the Highway Capacity Manual (HCM) [13]. It represents a quality measurement used to describe the percentage of operational conditions within a traffic flow, defining six different levels of service, with LOS A representing free-flow conditions and LOS F representing high congested level. The floating content provides information about the traffic condition of its critical area, thus vehicles can only detect congested roads and compute alternative route within this area. Therefore, each vehicle looks just the subset of its route R restrained in the information provided by the floating content. Let $R' = \{e_{ij}, \dots, e_{xy}\} \mid R' \subseteq R \cap E(G')$ is this subset. First a vehicle needs to check if its current route will pass through a congested road, verifying if a road $e_{ij} \in R'$ has weight greater than a threshold ε that indicates some congestion. This verification is given by the function $\sigma(n)$

$$\sigma(n) = \begin{cases} 1 & \exists e_{ij} \in R' \mid w_{ij} \geq \varepsilon \\ 0 & \text{Otherwise} \end{cases}$$

where, the value 1 means that a vehicle has road e_{ij} in its current route R' which has weight w_{ij} up to the defined threshold and 0 otherwise.

Whenever a vehicle goes through a congested road an alternative route must be computed. Therefore, the vehicle computes this alternative route $\delta(e_{ij}, e_{xy})$ based on its current position and the last road $e_{ij} \in R'$ using the Probabilistic k-Shortest Path (PkSP) based on the road weights. PkSP uses the Boltzmann probabilistic algorithm [14] to select a route among the ones in a set of k -possible routes, to reduce the probability of shifting the congestion to another area. All things considered, it is important to notice that the entire alternative route computed is given by

$$R_{new} = \delta(e_{ij}, e_{xy}) \cup R' \setminus R$$

where, $\delta(e_{ij}, e_{xy})$ is the alternative route computed by the vehicle from its current position and the last road $e_{ij} \in R'$ and $R' \setminus R$ is the remaining of its original route.

IV. PERFORMANCE ANALYSIS

This section describes the performance evaluation of PANDORA. First of all, Subsection IV-A introduces the simula-

tion tools as well as the scenario used in our simulations. Subsections IV-B and IV-C present the network performance evaluation and traffic efficiency results, respectively. First, we analyze the network cost of our solution by evaluating the aggregation mechanism and the opportunistic content sharing. Then, we compare PANDORA with some ATMS solutions.

A. Methodology

For the simulations, we use the network simulator OM-NeT++ 4.3¹ and the framework SUMO (Simulator for Urban-MOBility)², version 0.25.0, to manage the scenario and the mobility of vehicles. We obtained realistic map scenarios from the OpenStreetMap tool. For the vehicular network, we used the framework Veins 4.3³ that implements the IEEE 802.11p and the signal attenuation model considering obstacles.

We used a Manhattan grid scenario with size of 16 km², which is composed by several blocks where vehicles can move in both directions. Furthermore, to ensure close to real overtaking, their speeds range from 0 to the maximum speed allowed on the road. In addition, to provide different traffic conditions, we used densities from 700 to 1500 vehicles/km². To ensure a critical area, we force traffic congestion in the center of the scenario. Therefore, we set up a single critical area placed in the center of the scenario, with a size of 2 km². In addition, we set the $r_a = 1$ km, the $r_f = 1.5$ km and $\varepsilon = 0.5$. Finally, as the main parameters, we set the bitrate to 18 Mbit/s at the MAC layer and the transmission power to 2.2 mW, resulting in a communication range of approximately 300 m under a two-ray ground propagation model.

B. Network evaluation

As we did not find any other fully-distributed solution in the literature that builds precise knowledge about traffic conditions, we compared PANDORA with the Baseline [1], in which all vehicles spread their information to all vehicles in the network using a robust and efficient data dissemination protocol proposed in [1]. The assessed metrics for the network performance evaluation were: *Coverage*, percentage of delivery messages to vehicles which have to receive; *Transmitted messages*, total number of transmitted messages to deliver the content; *Packet collisions*, total number of packet collisions during the transmissions; and *Delay*, average time spent to deliver all messages.

Figure 3 shows the results for all assessed networks metrics as a function of the density of vehicles. Figure 3(a) shows the coverage results. In the Baseline solution, every vehicle needs to spread its information to all vehicles in the network to build knowledge about the traffic condition, however, due to several transmissions (most of them redundant) in a short period of time, it overloads the network, consequently, many messages will not be delivered to all vehicles drastically decreasing the coverage. As can be seen, the Baseline coverage is only 60% when the density is 700 vehicles/km². The greater the density, the lower is the coverage, since the network will become more and more overloaded due to the increase in the number of transmissions (see Figure 3(b)). As result of that, Baseline reduce its coverage in up to 30%. Differently, PANDORA does not overload the network, since it creates knowledge about the traffic condition just to specific areas and forwards knowledge just to the vehicles that have interest on it. Moreover, by

¹<http://omnetpp.org/>

²<http://sumo.dlr.de/>

³<http://veins.car2x.org>

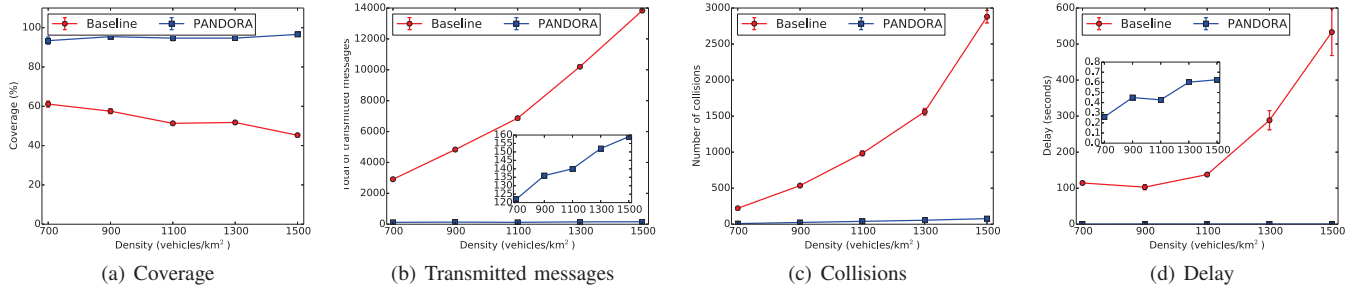


Fig. 3. Network evaluation results.

employing its floating content approach, information is not spread through the network. Actually, it is bounded to an AZ and it acts as a location-based service for delivering information just to vehicles within the AZ or to vehicles that will pass through this AZ. In this way, PANDORA generates fewer messages than the Baseline (see Figure 3(b)), providing a better coverage which is larger than 95% for all densities.

Figures 3(b) and 3(c) show the total number of messages transmitted and the number of packet collisions during the transmissions, respectively. These results describe how overloaded is the network in the Baseline solution. The high number of transmissions in a short period of time increases the probability of packet collision. However, whenever a packet collision occurs, the vehicles involved will try to transmit their messages at least once again, creating duplicated messages, thus, over increasing the number of transmitted messages and overloading further the network. In this way, the efficiency of the opportunistic content sharing employed in PANDORA shows up, since it does not waste bandwidth with unnecessary transmissions, and also just deliver the information to vehicles that really need to receive it. In addition, PANDORA reduces the number of transmitted messages in 90% and the number collisions in 95% compared with Baseline.

Finally, Figure 3(d) presents the average delay to deliver messages, as well as the consequence of the network overload produced by the Baseline. Such overload degrades the overall system performance, since it introduces undesired overhead which can generate negative impacts. Vehicles may not receive information about congested roads, thus getting stuck in a congested road, or they may receive this information late and detect that a road is congested when it is actually free. Because of that, the Baseline presents a delay greater than 100 seconds, while PANDORA presents average delay lower than 1 second. The overhead presented by the Baseline shows that it is not a suitable solution to deliver time sensitive information.

C. Traffic Efficiency evaluation

In this section, we evaluate the traffic flow of PANDORA compared with three other solutions UCONDES [4], SCORPION [6] and the solution proposed by Brennand et al. [5]. Additionally, we employ the original mobility, named Original Vehicular Mobility Traffic (OVMT) in the comparison. The assessed metrics for the vehicular traffic efficiency were: *Travel time*, average time spent to all vehicles to conclude their route; *Congestion time*, average time that all vehicles stay stopped in some traffic congestion; *Traveled distance*, average distance traveled by all vehicles; and *Average speed*, average speed of all vehicles during their entire route.

Figure 4 shows the results for all vehicular traffic efficiency metrics. In particular, Figure 4(a) shows the travel time results.

It is important to note that OVMT represents the original mobility (i.e. it does not alternate any route of all vehicles). Therefore, the larger the increase in density, the longer is the travel time, reaching approximately 28 minutes with density of 1500 vehicles/km². In addition, vehicles in OVMT spend approximately 68% of their entire travel time stuck in congestion (see Figure 4(b)). As the vehicles do not change their routes, OVMT presents the lowest traveled distance of approximately 5 km (see Figure 4(c)). As a consequence, due to long time spent in congestion, it presents an average speed lower than 15 km/h (see Figure 4(d)).

UCONDES increases the travelled distance approximately 15% as consequence of the congestion avoidance mechanism employed, which does not have any knowledge about the traffic condition. Hence, UCONDES can create congestion in other areas and it will have to avoid them as well, thus increasing the traveled distance (see Figure 4(c)). However, UCONDES decreases the congestion time in approximately 25% compared to OVMT when the density is 1500 vehicles/km² (see Figure 4(b)), increasing the distance in approximately 8% (see Figure 4(c)). As a consequence of the congestion time, UCONDES presents an average speed of approximately 12 km/h (see Figure 4(d)).

Differently, SCORPION and Brennand et al. proposal, use a predefined interval to detect and control traffic congestion. In this way, they decrease the travel time in approximately 30% and 15% compared to OVMT (see Figure 4(c)) when density is 1500 vehicles/km², respectively. Moreover, SCORPION presents a reduction in congestion time of 50%, while Brennand et al. proposal present a reduction of 30% (see Figure 4(b)). The great reduction presented in SCORPION relies to its cooperative rerouting, where it can make a better traffic distribution, controlling and reducing traffic congestions (see Figure 4(a) and Figure 4(b)). On the other hand, both SCORPION and Brennand et al. proposal increase the traveled distance (see Figure 4(c)). Such increase is due to the rerouting mechanism employed by them. However, in the approach proposed by Brennand et al. due to high congestion time, vehicles need to compute greater routes when compared to SCORPION to avoid congestion. Such problem is better addressed by SCORPION, since it can improve the traffic efficiency by increasing only 10% of the traveled distance when compared to OVMT, while the proposal of Brennand et al. proposal increases approximately 20% (see Figure 4(c)). Both solutions present a similar average speed of approximately 13 km/h (see Figure 4(d)). Despite SCORPION presenting a better traffic congestion control, the solution proposed by Brennand et al. proposal reaches a similar average speed due to long distance traveled by vehicles.

Finally, PANDORA presents the lowest travel time and

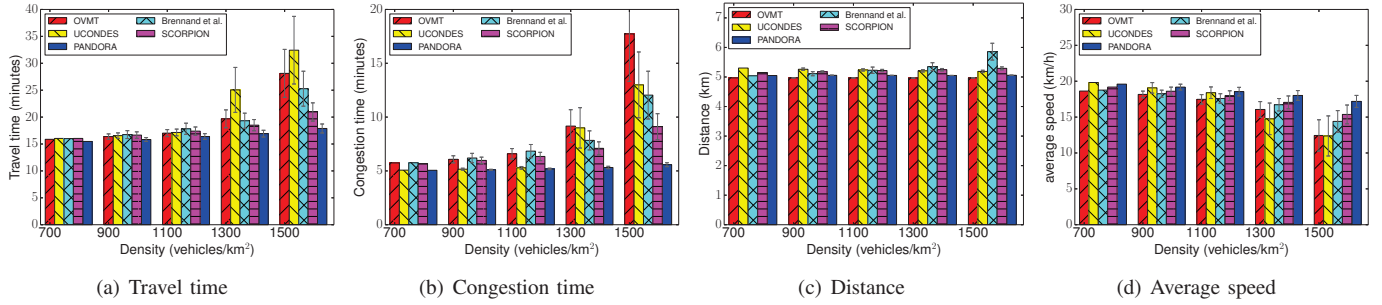


Fig. 4. Traffic efficiency evaluation results.

congestion time, decreasing approximately 46% and 72%, compared to OVMT, 53% and 61% compared to UCONDES, 40% and 55% compared Brennan et al. and 20% and 45% compared to SCORPION (see Figure 4(b) and Figure 4(a)). Such reduction is the result of the precise information about the traffic condition delivered to the vehicles in critical areas jointly with the efficient congestion detection and route guidance mechanism employed. However, despite PANDORA increasing the travel distance (Figure 4(c)) it presents the highest average speed which is up to 17 km/h (see Figure 4(d)).

Thanks to the low overhead, PANDORA can update the knowledge about the traffic condition in critical areas potentially on-the-fly. Thus, it can detect congestion first than the other solutions [5], [6]. Such low overhead is the result of PANDORA fully-distributed approach in which precise knowledge about the traffic condition within critical areas can be built, making each vehicle to detect congestion and compute alternative routes to itself. The overhead introduced by SCORPION and the solution proposed by Brennan et al. is higher due to their centralized approach. Moreover, the accuracy of the knowledge about the traffic condition depends on the frequency of data collection. As a consequence, the number of vehicles stuck in congestion also increases.

V. CONCLUSION

Traffic congestion has become a daily problem affecting several aspects of modern society. Centralized ATMS present several characteristics which can degrade performance. In this work, we present PANDORA, a fully-distributed ATMS based on opportunistic content sharing to improve the overall vehicle traffic efficiency, which provides precise knowledge about the traffic condition of critical areas. PANDORA was extensively compared to other known solutions using metrics such as travel time, congestion time, traveled distance, average speed and several network metrics. Results indicate that PANDORA outperforms the assessed solutions in different scenarios and different key requirements of ATMS. As future work, we intend to analyze PANDORA in more realistic scenarios, using real environments mobility traces. In addition, we intend to explore how the size of AZ impacts on the overall performance.

VI. ACKNOWLEDGMENTS

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